FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

ITDS 294
Fundamentals of Electronic Commerce Systems

Session 4 - 5 : Building an E-Commerce Presence

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Building an E-Commerce Presence →

กระบวนการในการสร้าง E-Commerce การเลือกซอฟต์แวร์ ประเด็นต่าง ๆ ที่น่าสนใจในการสร้าง E-Commerce รูปแบบของ Mobile Commerce Platform การออกแบบและประเด็นที่ต้องพิจารณา

Case Study : Skyscanner - The One-Stop Travel Platform →

เรียนรู้ E-Commence Presence ของ Skyscanner และการใช้ประโยชน์ของ E-Commerce ในการแข่งขันกับคู่แข่งธุรกิจ

Learning Outcome for Session 4 - 5

Building an E-Commerce Presence

นิสิตสามารถอธิบายกระบวนการในการสร้าง E-Commerce และเลือกรูปแบบที่เหมาะสมในการนำไปใช้งาน

Case Study :
Skyscanner - The One-Stop Travel Platform

นิสิตสามารถประยุกต์ความรู้ที่ได้จากกรณีศึกษาไปต่อยอดใช้สำหรับวางแผนการสร้างธุรกิจพาณิชย์อิเล็กทรอนิกส์ในอนาคตได้

Building an E-Commerce Presence

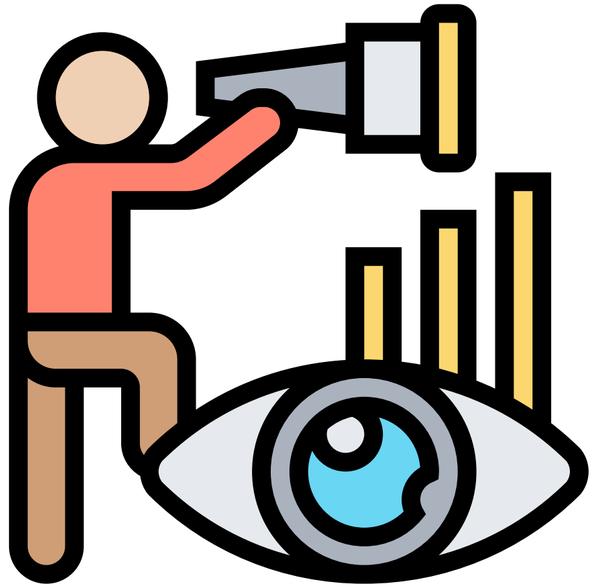
กระบวนการในการสร้าง E-Commerce การเลือกซอฟต์แวร์ ประเด็นต่าง ๆ ที่น่าสนใจในการสร้าง E-Commerce รูปแบบของ Mobile Commerce Platform การออกแบบและประเด็นที่ต้องพิจารณา

Imagine : Your E-Commerce Presence

Before you begin to build a website or app of your own, there are some important questions you will need to think about and answer. The answers to these questions will drive the development and implementation of your e-commerce presence.

What's The Idea? : The Visioning Process

Before you can plan and actually build an e-commerce presence, you need to have a vision of what you hope to accomplish and how you hope to accomplish it. The vision includes not just a statement of mission, but also identification of the target audience, characterization of the market space, a strategic analysis, a marketing matrix, and a development timeline. It starts with a dream of what’s possible and concludes with a timeline and preliminary budget for development.



Where's the Money? : Business and Revenue Model

Once you have defined a mission statement, a vision, you need to start thinking about where the money will be coming from. You will need to develop a preliminary idea of your business and revenue models. You don’t need detailed revenue and cost projections at this point. Instead, you need a general idea of how your business will generate revenues.

Who and Where is the Target Audience?

Without a clear understanding of your target audience, you will not have a successful e-commerce presence. There are two questions here: who is your target audience and where can you best reach them? Your target audience can be described in a number of ways: demographics, behavior patterns (lifestyle), current consumption patterns (online vs. offline purchasing), digital usage patterns, content creation preferences (blogs, social networks, sites like Pinterest), and buyer personas (profiles of your typical customer).

Understanding the demographics of your target audience is usually the first step. Demographic information includes age, income, gender, and location. In some cases, this may be obvious, and in others, much less so.

What is the Ballpark? Characterize the Marketplace

The chances of your success will depend greatly on the characteristics of the market you are about to enter, and not just on your entrepreneurial brilliance. Enter a declining market filled with strong competitors, and you will multiply your chances of failure. Enter a market that is emerging, growing, and has few competitors, and you stand a better chance. Enter a market where there are no players, and you will either be rewarded handsomely with a profitable monopoly on a successful product no one else thought of (Apple) or you will be quickly forgotten because there isn't a market for your product at this point in time

What is the Ballpark? Characterize the Marketplace

In addition, you will want to know about the structure of the market: competitors and substitute products.

What are the features of the marketplace you are about to enter? Is the market growing, or receding in size? If it's growing, among which age and income groups? Is the marketplace shifting from offline to online delivery? If so, is the market moving toward traditional websites or the mobile platform? Is there a special role for a mobile presence in this market? What percentage of your target audience uses a website, smartphone, or tablet? What about social networks? What's the buzz on products like yours? Are your potential customers talking about the products and services you want to offer on Facebook, Twitter, Pinterest, Instagram, or blogs? How many blogs focus on products like yours? How many Twitter posts mention similar offerings? How many Facebook Likes (signs of customer engagement) are attached to products you want to offer?

Where's the Content Coming From?

Websites are like books: they’re composed of a lot of pages that have content ranging from text, to graphics, photos, and videos. This content is what search engines catalog as they crawl through all the new and changed web pages on the Internet. The content is why your customers visit your site and either purchase things or look at ads that gener- ate revenue for you. Therefore, the content is the single most important foundation for your revenue and ultimate success.

กระดานกลยุทธ์ก่อนเริ่มสร้างระบบ (The Strategic Canvas)

วิสัยทัศน์ & รายได้ (Vision & Revenue)

พันธกิจของธุรกิจคืออะไร? (Mission Statement)

โมเดลธุรกิจและที่มาของรายได้ (Business & Revenue Model)

กลุ่มเป้าหมาย (Target Audience)

ข้อมูลประชากร (Demographics: อายุ, รายได้, เพศ)

พฤติกรรมออนไลน์และออฟไลน์ (Behavior & Digital Usage)

พื้นที่ตลาด (Market Space)

แนวโน้มอุตสาหกรรม (ตลาดกำลังโตหรือหดตัว?)

คู่แข่งและสินค้าทดแทน (Competitors & Substitutes)

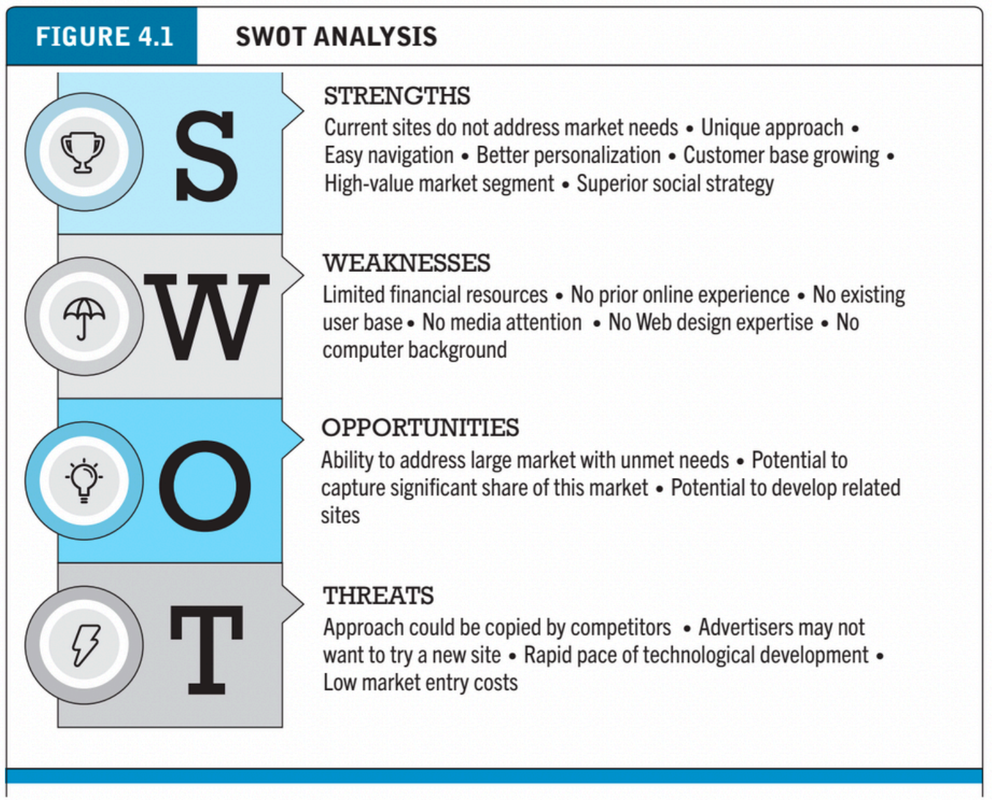
คอนเทนต์ (The Content Engine)

รากฐานสำคัญที่สุดของรายได้ (The single most important foundation for revenue)

ข้อความ, รูปภาพ, วิดีโอ ที่ดึงดูดลูกค้าและ Search Engine

Know Yourself : Conduct a SWOT Analysis

A SWOT analysis is a simple but powerful method for strategizing about your business and understanding where you should focus your efforts. In a SWOT analysis, you describe your strengths, weaknesses, opportunities, and threats (SWOT).



A SWOT analysis describes your firm's strengths, weaknesses, opportunities, and threats

รู้จักตัวเองด้วยตารางวิเคราะห์ SWOT (Self-Assessment)

Strengths – จุดแข็ง

สินค้าตอบโจทย์ตลาด, การนำทางเว็บง่าย, การทำ Personalization ที่ดีกว่า

Weaknesses – จุดอ่อน

ทุนจำกัด, ขาดประสบการณ์ออนไลน์, ไม่มีฐานผู้ใช้เดิม, ขาดความเชี่ยวชาญด้านเว็บ

Opportunities – โอกาส

ตลาดขนาดใหญ่ที่ยังไม่มีใครตอบสนอง, โอกาสครองส่วนแบ่งตลาด, การขยายเว็บไซต์ที่เกี่ยวข้อง

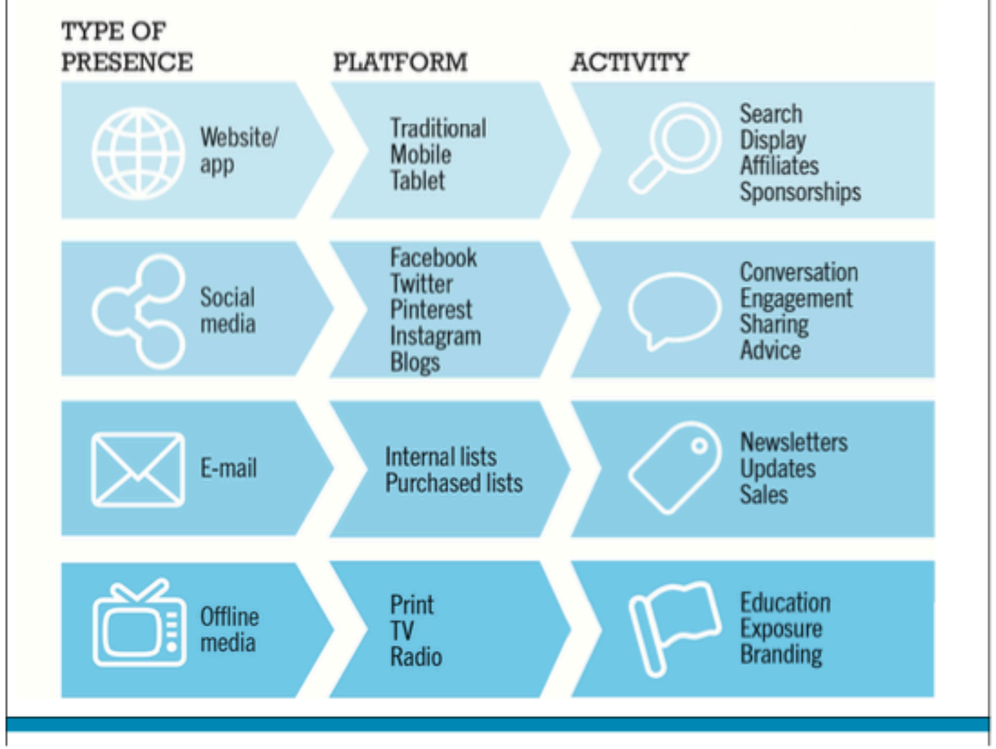
Threats – อุปสรรค

คู่แข่งลอกเลียนแบบ, ผู้ลงโฆษณาไม่กล้าเสี่ยงกับเว็บใหม่, เทคโนโลยีเปลี่ยนเร็ว, ต้นทุนการเข้าสู่ตลาดสู่ตลาดต่ำ (Low entry barriers)

Develop an E-Commerce Presence Map

E-commerce has moved from being a desktop-centric activity to a mobile and tablet- based activity as well. The following figure provides a roadmap to the platforms and related activities you will need to think about when developing your e-commerce presence.

FIGURE 4.2 E-COMMERCE PRESENCE MAP



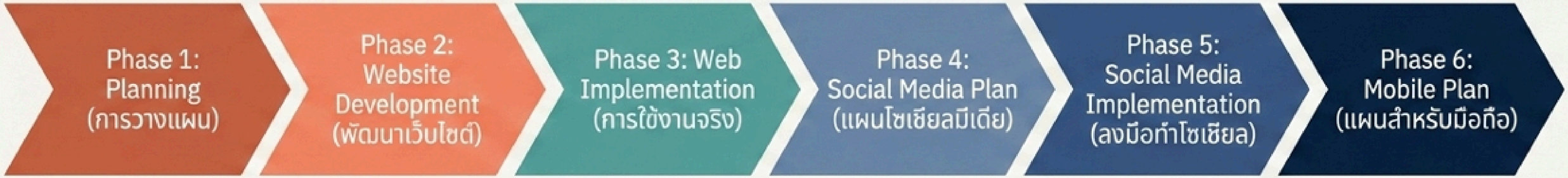
Develop a Timeline : Presence Milestones

It's a good idea for you to have a rough idea of the time frame for developing your e-commerce presence when you begin. You should break your project down into a small number of phases that could be completed within a specified time. Six phases are usually enough detail at this point. The following table illustrates a one-year timeline for the development of a startup e-commerce company.

TABLE 4.1 E-COMMERCE PRESENCE TIMELINE

PHASE	ACTIVITY	MILESTONE
Phase 1: Planning	Envision e-commerce presence; determine personnel	Mission statement
Phase 2: Website development	Acquire content; develop a site design; arrange for hosting the site	Website plan
Phase 3: Web implementation	Develop keywords and metatags; focus on search engine optimization; identify potential sponsors	A functional website
Phase 4: Social media plan	Identify appropriate social platforms and content for your products and services	A social media plan
Phase 5: Social media implementation	Develop Facebook, Instagram, Twitter, and Pinterest presence	Functioning social media presence
Phase 6: Mobile plan	Develop a mobile plan; consider options for porting your website to mobile devices	A mobile plan

แผนการดำเนินงาน 1 ปี (Presence Timeline)



Milestone: คำหนดพันธกิจ (Mission statement)

Milestone: แผนผังเว็บและการโฮสต์ (Website plan)

Milestone: เว็บไซต์ที่ใช้งานได้จริง (A functional website)

Milestone: แผนกลยุทธ์โซเชียล (A social media plan)

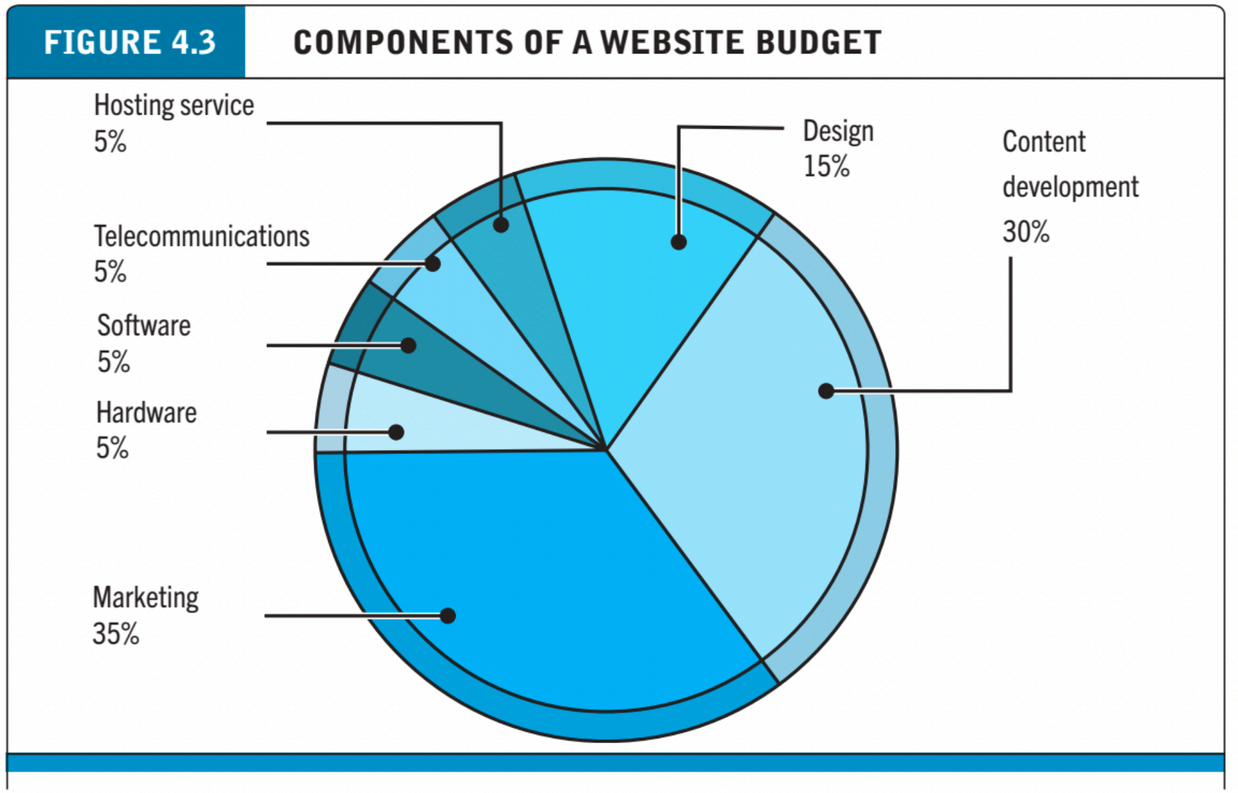
Milestone: ตัวตนบนโซเชียลที่สมบูรณ์ (Functioning presence)

Milestone: แผนงานระบบมือถือ (A mobile plan)

How much will this cost?

It’s too early in the process to develop a detailed budget for your e-commerce presence, but it is a good time to develop a preliminary idea of the costs involved.

FIGURE 4.3 **COMPONENTS OF A WEBSITE BUDGET**



While hardware and software costs have fallen dramatically, websites have significant design, content development, and marketing costs.

Building an E-Commerce Presence : A Systematic Approach

Once you have developed a vision of the e-commerce presence you want to build, it’s time to start thinking about how to build and implement that presence.

Building a successful e-commerce presence requires a keen understanding of business, technology, and social issues, as well as a systematic approach. E-commerce is just too important to be left totally to technologists and programmers.

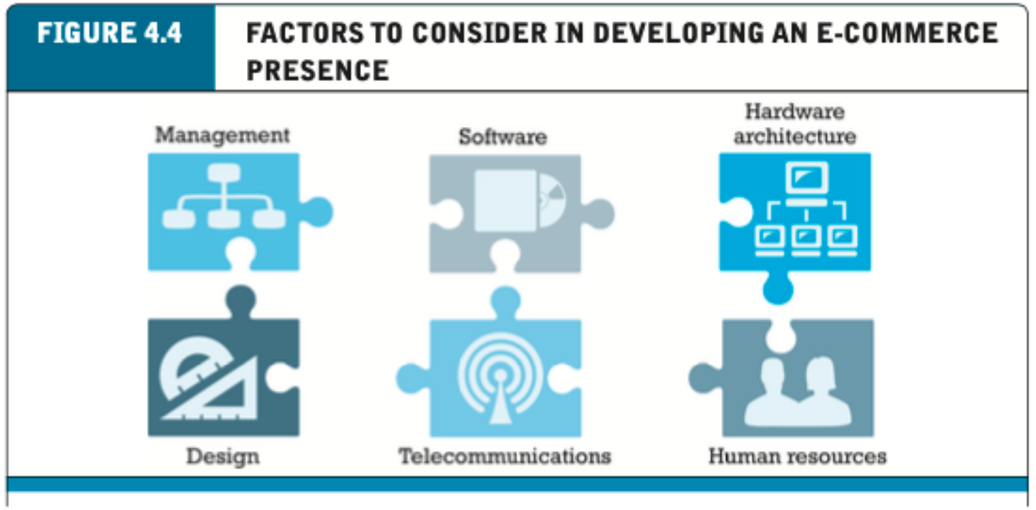
Building an E-Commerce Presence : A Systematic Approach

The two most important management challenges are (1) developing a clear understanding of your business objectives and (2) knowing how to choose the right technology to achieve those objectives. The first challenge requires you to build a plan for develop- ing your firm's presence. The second challenge requires you to understand some of the basic elements of e-commerce infrastructure.

Even if you decide to outsource the development effort and operation to a service provider, you will still need to have a development plan and some understanding of the basic e-commerce infrastructure issues such as cost, capability, and constraints. Without a plan and a knowledge base, you will not be able to make sound management decisions about e-commerce within your firm.

Let the business drive the technology

Building an E-Commerce Presence : A Systematic Approach



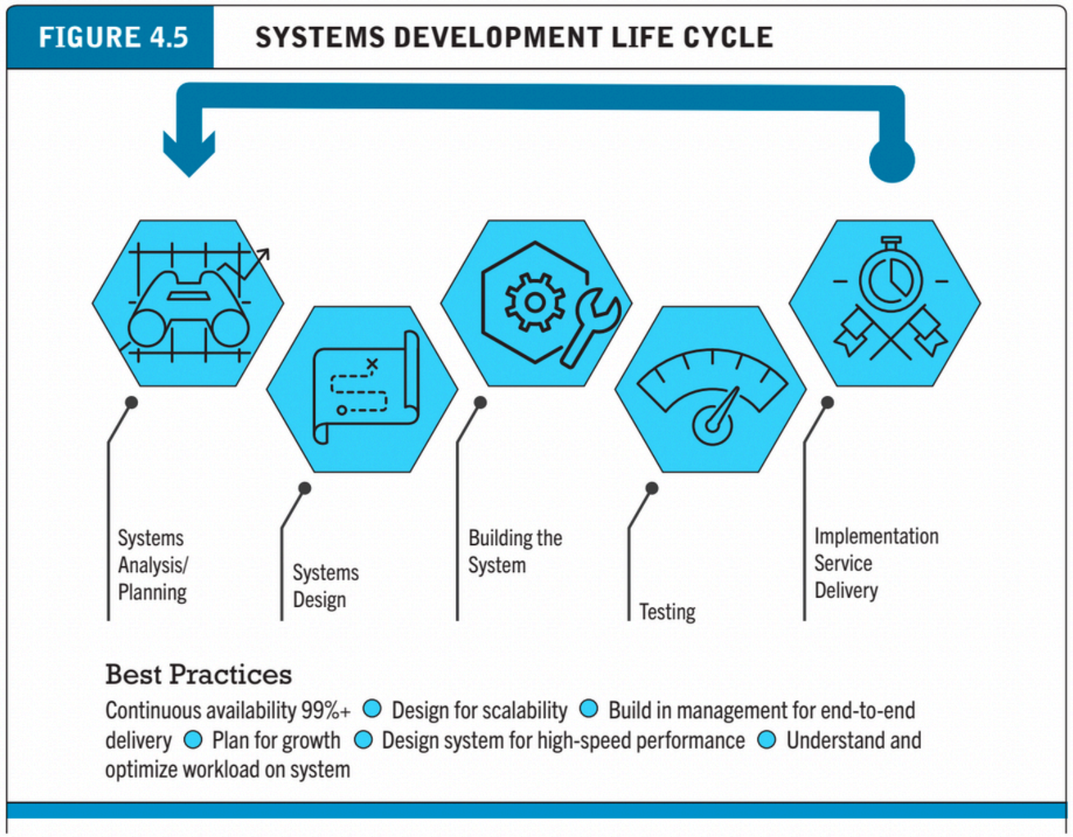
First, you must be aware of the main areas where you will need to make decisions. On the organizational and human resources fronts, you will have to bring together a team of individuals who possess the skill sets needed to build and man- age a successful e-commerce presence. This team will make the key decisions about business objectives and strategy, technology, design, and social and information policies. The entire development effort must be closely managed if you hope to avoid the disasters that have occurred at some firms.

Building an E-Commerce Presence : A Systematic Approach

You will also need to make decisions about hardware, software, and telecommunications infrastructure. The demands of your customers should drive your choices of technology. Your customers will want technology that enables them to find what they want easily, view the product, purchase the product, and then receive the product from your warehouses quickly. You will also have to carefully consider design. Once you have identified the key decision areas, you will need to think about a plan for developing the project. There are a number of different methodologies for building information systems such as websites. One of the most traditional methods is the **systems development life cycle**.

System Development Life Cycle

The **systems development life cycle** (**SDLC**) is a methodology for understanding the business objectives of any system and designing an appropriate solution.



The systems development life cycle (SDLC) has five major steps: systems analysis/planning, systems design, building the system, testing, and implementation and maintenance.

System Development Life Cycle

Systems Analysis/Planning: Identify Business Objectives, System Functionality, and Information Requirements

- In the systems analysis/planning step of the SDLC, you try to answer the question, “What do we want this e-commerce site or app to do for our business?” The key point is to let the business decisions drive the technology, not the reverse. This will ensure that your technology platform is aligned with your business.
- One way to start is to identify the specific business objectives for your site, and then develop a list of system functionalities and information requirements. Business objectives are simply capabilities you want your site to have.
- **System functionalities** are types of information systems capabilities you will need to achieve your business objectives. The **information requirements** for a system are the information elements that the system must produce in order to achieve the business objectives. You will need to provide these lists to system developers and programmers so they know what you as the manager expect them to do.

System Development Life Cycle

TABLE 4.2 SYSTEM ANALYSIS: BUSINESS OBJECTIVES, SYSTEM FUNCTIONALITIES, AND INFORMATION REQUIREMENTS FOR A TYPICAL E-COMMERCE WEBSITE

BUSINESS OBJECTIVE	SYSTEM FUNCTIONALITY	INFORMATION REQUIREMENTS
Display goods.	Digital catalog	Dynamic text and graphics
Provide product information (content).	Product database	Product description, stocking numbers, inventory levels
Personalize/customize product.	Customer on-site tracking	Website log for every customer visit; data-mining capability to identify common customer paths and appropriate responses
Engage customers in conversations.	On-site blog; user forums	Customer comments captured by software with blogging and community forum functionality
Execute a transaction.	Shopping cart/payment system	Secure credit card clearing; multiple payment options
Accumulate customer information.	Customer database	Name, address, phone, and e-mail for all customers; online customer registration
Provide after-sale customer support.	Sales database	Customer ID, product, date, payment, shipment date
Coordinate marketing/advertising.	Ad server, e-mail server, e-mail, campaign manager, ad banner manager	Site behavior log of prospects and customers linked to e-mail and banner ad campaigns
Understand marketing effectiveness.	Site tracking and reporting system	Number of unique visitors, pages visited, and products purchased, identified by marketing campaign
Provide production and supplier links.	Inventory management system	Product and inventory levels, supplier ID and contact, order quantity data by product

System Development Life Cycle

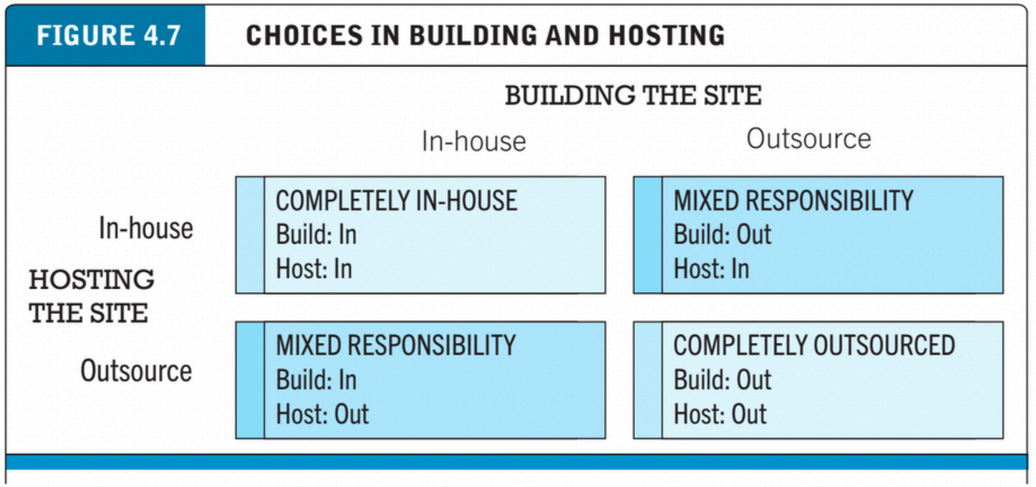
System Design: Hardware and Software Platforms

- To consider just how all this functionality will be delivered, you must come up with a system design specification—a description of the main components in the system and their relationship to one another. The system design itself can be broken down into two components: a logical design and a physical design.
- A **logical design** includes a data flow diagram that describes the flow of information at your e-commerce site, the processing functions that must be performed, and the databases that will be used. The logical design also includes a description of the security and emergency backup procedures that will be instituted, and the controls that will be used in the system.
- A **physical design** translates the logical design into physical components. For instance, the physical design details the specific model of server to be purchased, the software to be used, the size of the telecommunications link that will be required, the way the system will be backed up and protected from outsiders, and so on.

System Development Life Cycle

Building the System: In-house Versus Outsourcing

- You have many choices, and much depends on the amount of money you are willing to spend. Choices range from outsourcing everything (including the actual systems analysis and design) to building everything yourself (in-house).
- **Outsourcing** means that you will hire an outside vendor to provide the services involved in building the site rather than using in-house personnel. You also have a second decision to make: will you host (operate) the site on your firm's own servers or will you outsource the hosting to a web host provider?



You have a number of alternatives to consider when building and hosting an e-commerce site.

System Development Life Cycle

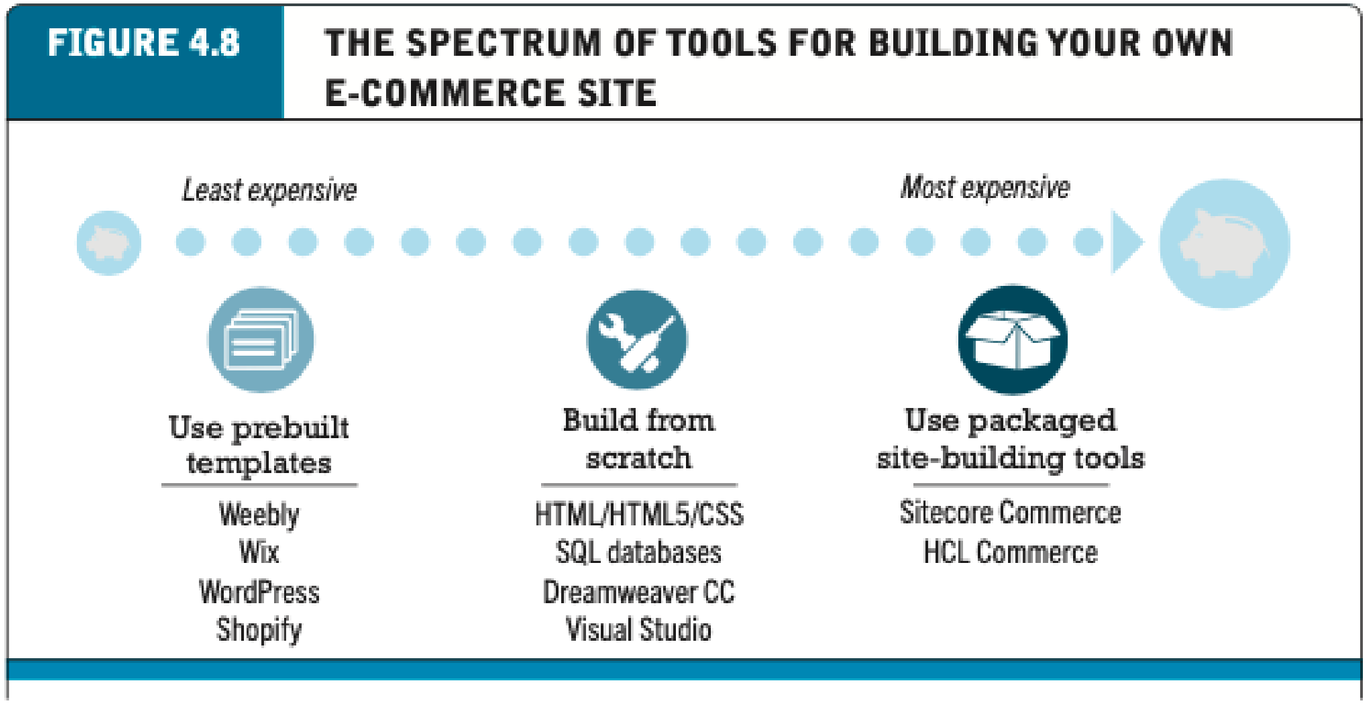
Building the System: In-house Versus Outsourcing

- Let’s take the building decision first. If you elect to build your own site, there are a range of options. Unless you are fairly skilled, you should use a pre-built template to create the website.
- One of the least expensive and most widely used site building tools is WordPress. WordPress is a website development tool with a sophisticated content management system. A **content management system** (**CMS**) is a database software program specifically designed to manage structured and unstructured data and objects in a website environment. A CMS provides web managers and designers with a centralized control structure to manage website content. WordPress also has thousands of widgets and plug-ins, such as WooCommerce, which provides an e-commerce platform, that can extend the functionality of a website. Other similar website building tools are provided by Weebly, Wix, Squarespace, and Shopify.
- If you want more customization than using a pre-built template can provide and have some programming experience, you can build the site yourself.

System Development Life Cycle

Building the System: In-house Versus Outsourcing

- If you choose more expensive site-building packages, you will be purchasing state- of-the art software that is well tested. You could get to market sooner.



System Development Life Cycle

Building the System: In-house Versus Outsourcing

- **Host Your Own versus Outsourcing**
  - Most businesses choose to outsource **hosting** and pay a company to host their website, which means that the hosting company is responsible for ensuring the site is "live," or accessible, 24 hours a day.
  - You can also choose to *co-locate*. With a **co-location** agreement, your firm purchases or leases a web server (and has total control over its operation) but locates the server in a vendor's physical facility. The vendor maintains the facility, communications lines, and the machinery.
  - **Cloud services** are rapidly replacing co-location because they are less expensive and arguably more reliable. Unlike with co-location, your firm does not own the hard- ware. Cloud service providers offer a standardized infrastructure, virtualization technology, and employ a pay-as-you-go billing system

TABLE 4.3 KEY PLAYERS: HOSTING/CO-LOCATION/CLOUD SERVICES

Alibaba Cloud	IBM Cloud
Amazon Web Services (AWS) EC2	Liquid Web
CenturyLink	Microsoft Azure
Digital Realty Trust	OVHcloud
Fujitsu	Rackspace
Google Cloud	Virtualstream

System Development Life Cycle

Testing the System

- Once the system has been built and programmed, you will have to engage in a testing process. Depending on the size of the system, this could be fairly difficult and lengthy. Testing is required whether the system is outsourced or built in-house.
- **Unit testing** involves testing the site's program modules one at a time.
- **System testing** involves testing the site as a whole, in the same way a typical user would when using the site.
- Final **acceptance testing** requires that the firm's key personnel and managers in marketing, production, sales, and general management actually use the system as installed on a test Internet or intranet server. This acceptance test verifies that the business objectives of the system as originally conceived are in fact working.
- Another form of testing is called **A/B testing** (or **split testing**). This form of testing involves showing two versions (A and B) of a web page or website to different users to see which one performs better. There are several different types of A/B testing that can be used for a website design project.

System Development Life Cycle

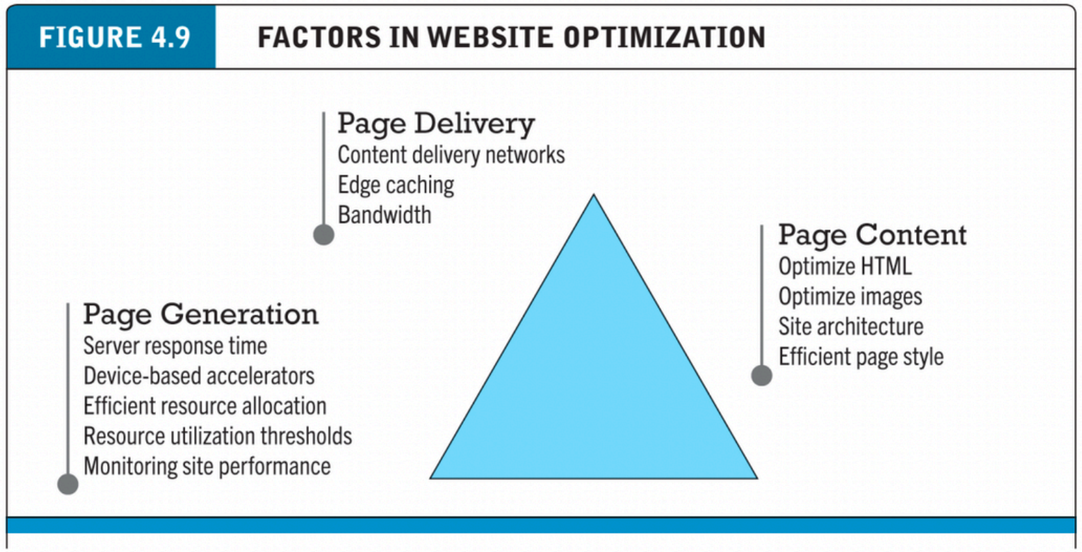
Implementation, Maintenance, and Optimization

- Most people unfamiliar with systems erroneously think that once an information system is installed, the process is over. In fact, while the beginning of the process is over, the operational life of a system is just beginning. Systems break down for a variety of reasons—most of them unpredictable. Therefore, they need continual checking, testing, and repair. **Systems maintenance** is vital, but sometimes not budgeted for. In general, the annual system maintenance cost will roughly parallel the development cost.
- The long-term success of an e-commerce site will depend on a dedicated team of employees (the web team) whose sole job is to monitor and adapt the site to changing market conditions. The web team must be multi-skilled; it will typically include programmers, designers, and business managers drawn from marketing, production, and sales support.
- One of the first tasks of the web team is to listen to customers' feedback on the site and respond to that feedback as necessary. A second task is to develop a systematic monitoring and testing plan to be followed weekly to ensure all the links are operating, prices are correct, and pages are updated. Other important tasks of the web team include **benchmarking** (a process in which the site is compared with those of competitors in terms of response speed, quality of layout, and design) and keeping the site current on pricing and promotions.

System Development Life Cycle

Implementation, Maintenance, and Optimization

- If you are developing your firm's website in-house, you will need to consider these issues. The purpose of a website is to deliver content to customers and to complete transactions. The faster and more reliably these two objectives are met, the more effective the website is from a commerce perspective.
- The optimization of website performance is more complicated than it seems and involves a number of factors, including page content, page generation, and page delivery



Website optimization requires that you consider three factors: page content, page generation, and page delivery.

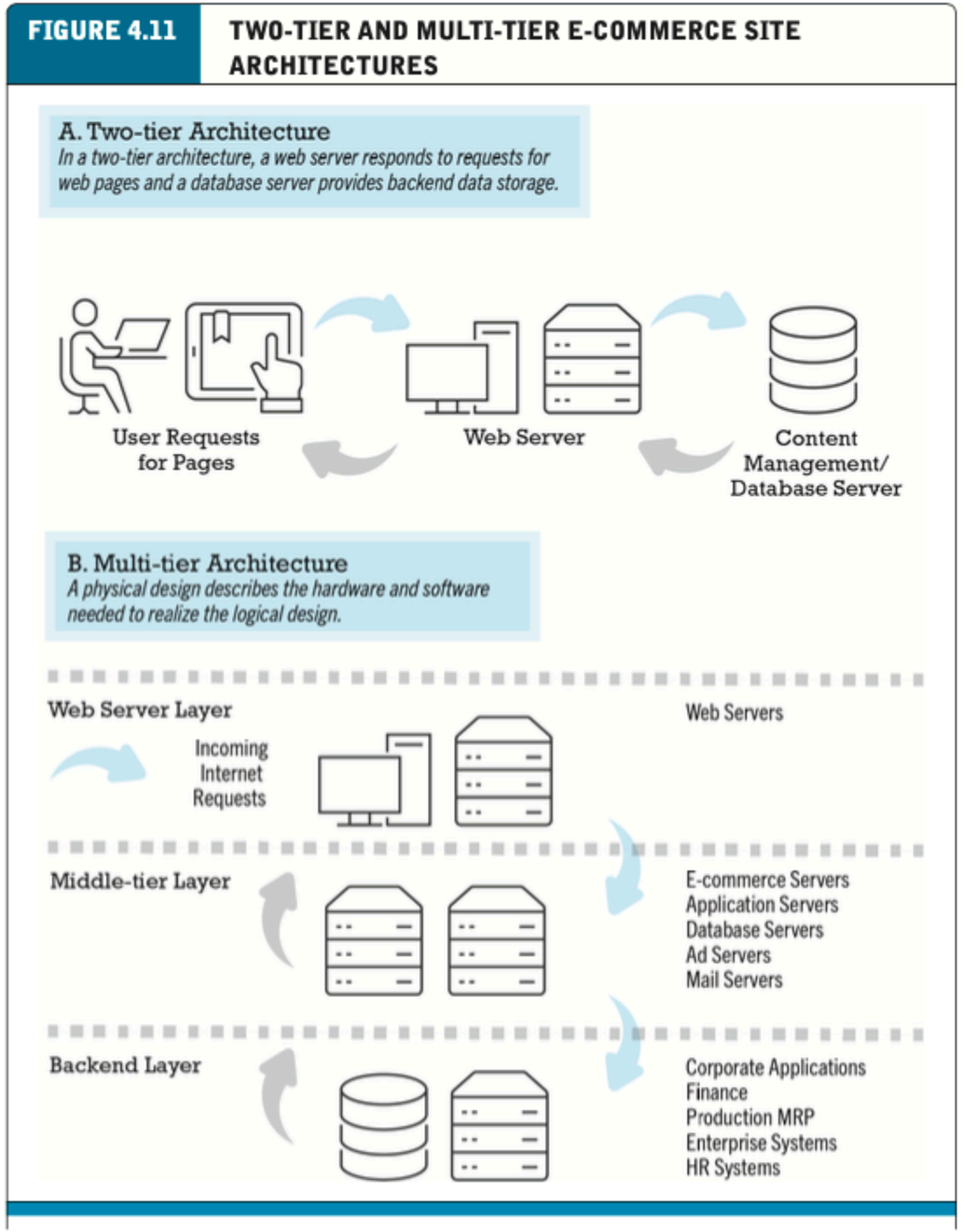
Choosing Software

Although today, many businesses choose to outsource their e-commerce infrastructure to cloud providers, it is still very important to have a basic understanding of the underlying software and hardware components that comprise that presence.

Choosing Software

Simple VS Multi-tiered Website Architecture

- System architecture refers to the arrangement of software, machinery, and tasks in an information system needed to achieve a specific functionality.
- In **two-tier architecture**, a web server responds to requests for web pages and a database server provides backend data storage.
- In a **multi-tier architecture**, in contrast, the web server is linked to a middle-tier layer that typically includes a series of application servers that perform specific tasks, as well as to a backend layer of existing corporate systems containing product, customer, and pricing information.



Choosing Software

Web Server Software

- When you choose web server software, you will also be choosing an operating system for your site's computers.
- Apache, which works with Linux and Unix operating systems, is the leading web server software in terms of domains, active sites, and usage by the top 1 million busiest sites.
- Microsoft Internet Information Services (IIS) is another popular type of web server software. IIS is based on the Windows operating system and is compatible with a wide selection of Microsoft utility and support programs.

Choosing Software

Application Servers

- **Web application servers** are software programs that provide the specific business func- tionality required of a website. The basic idea of application servers is to isolate the business applications from the details of displaying web pages to users on the front end and the details of connecting to databases on the back end. Application servers are a kind of middleware software that provides the glue connecting traditional corporate systems to the customer as well as all the functionality needed to conduct e-commerce.

Choosing Software

E-Commerce Software Platforms

- Rather than build your site from a collection of disparate software applications, it is easier, faster, and generally more cost-effective to purchase a **merchant server software package** (also called an **e-commerce software platform**). Merchant server software offers an integrated environment that promises to provide most or all of the functional- ity and capabilities you will need to develop a sophisticated, customer-centric site.

TABLE 4.4 OPEN-SOURCE SOFTWARE OPTIONS

FUNCTIONALITY	OPEN-SOURCE SOFTWARE
Web server	Apache (the leading web server for small and medium-sized businesses)
Shopping cart, online catalog	Many providers: osCommerce, Zen Cart, AgoraCart, X-cart, AspDotNetStorefront
Credit card processing	Credit card acceptance is typically provided in shopping cart software, but you may need a merchant account from a bank as well.
Database	MySQL (the leading open-source SQL database for businesses)
Programming/scripting language	PHP is a scripting language embedded in HTML documents but executed by the server, providing server-side execution with the simplicity of HTML editing. JavaScript programs are typically client-side programs that provide user interface components. Ruby on Rails (RoR, Rails) and Django are popular open-source web application frameworks. Python and Perl are two other open-source programming languages used in web development.
Analytics	Analytics keep track of your site's customer activities and the success of your web advertising campaign. You can also use Google Analytics if you advertise on Google, which provides good tracking tools; most hosting services will provide these services as well. Other open-source analytic tools include Matomo and Open Web Analytics.

Choosing Software

Choosing E-Commerce Software Platforms

The following are some of the key factors to consider:

- Functionality, including availability on an SaaS basis
- Support for different business models, including m-commerce
- Business process modeling tools
- Visual site management tools and reporting
- Performance and scalability
- Connectivity to existing business systems
- Compliance with standards
- Global and multicultural capability
- Local sales tax and shipping rules

Other Interesting E-Commerce Issues

Now that you understand the key factors affecting the speed, capacity, and scalability of your website, we can consider some other important requirements. You will need a coherent website design that makes business sense—not necessarily a site to wow visitors or excite them, but to sell them something.

E-Commerce Design Issues

E-Commerce Website Features that Annoy Customers

TABLE 4.8 E-COMMERCE WEBSITE FEATURES THAT ANNOY CUSTOMERS

- Requiring user to view ad or intro page before going to website content
- Pop-up and pop-under ads and windows
- Too many clicks to get to the content
- Links that don't work
- Confusing navigation; no search function
- Requirement to register and log in before viewing content or ordering
- Pages that load slowly
- Content that is out of date
- Inability to use browser's Back button
- No contact information available (web form only)
- Unnecessary splash screens, animation, etc.
- Music or other audio that plays automatically
- Unprofessional design elements
- Text not easily legible due to size, color, format
- Typographical errors
- No or unclear returns policy

E-Commerce Design Issues

Important Factors in Successful E-Commerce Site Design

TABLE 4.9 THE EIGHT MOST IMPORTANT FACTORS IN SUCCESSFUL E-COMMERCE SITE DESIGN

FACTOR	DESCRIPTION
Functionality	Pages that work, load quickly, and point the customer toward your product offerings
Informational	Links that customers can easily find to discover more about you and your products
Ease of use	Simple, foolproof navigation
Redundant navigation	Alternative navigation to the same content
Ease of purchase	One or two clicks to purchase
Multi-browser functionality	Site works with the most popular browsers
Simple graphics	Avoids distracting, obnoxious graphics and sounds that the user cannot control
Legible text	Avoids backgrounds that distort text or make it illegible

E-Commerce Personalization Issues

Personalization Tools

- **Personalization** (the ability to treat people based on their personal qualities and prior history with your site) and **Customization** (the ability to change the product to better fit the needs of the customer) are two key elements of e-commerce that potentially can make it nearly as powerful as a traditional marketplace, and perhaps even more powerful than direct mail or shopping at an anonymous suburban shopping mall.
- There are a number of methods for achieving personalization and customization. One basic method involves the placement of cookie files on the user's client computer. A cookie is a small text file placed on the user's client computer that can contain any kind of information about the customer, such as customer ID, campaign ID, or purchases at the site. When the user returns to the site, or indeed goes further into the site, the customer's prior history can be accessed from a database. Information gathered on prior visits can then be used to personalize the visit and customize the product.
- There are also a number of other more sophisticated tools available that offer intelligent customer segmentation, personalization, and individualized product recom- mendations. Examples include Montetate, Barilliance, and Evergage. Most e-commerce suites, such as Salesforce Commerce Cloud, include similar functionality. At the other end of the spectrum, Google Optimize is a free tool that can customize a website based on visitor location, browser device, and digital behavior.

Developing a Mobile Website and Building Mobile Applications

Today, building a website is just one part of developing an e-commerce presence. Given that 90% of all Internet users worldwide access the Web at least part of the time from mobile devices, businesses today need to develop mobile websites and mobile web apps, native apps, or hybrid apps, in order to interact with customers, suppliers, and employees. Deciding which of these extended web presence tools to use is a first step.

Developing a Mobile Website and Building Mobile Applications

M-Commerce Platform

- A **mobile website** is a version of a regular website that is scaled down in content and navigation so that users can find what they want and move quickly to a decision or purchase.
- A **mobile web app** is an application built to run on the mobile web browser built into a smartphone or tablet computer.
- A **native app** is an application designed specifically to operate using the mobile device's hardware and operating system.
- A **hybrid app** has many of the features of both a native app and a mobile web app. Like a native app, it runs inside a native container on the mobile device and has access to the device's APIs, enabling it to take advantage of many of the device's features, such as a gyroscope, that are normally not accessible by a mobile web app.

Developing a Mobile Website and Building Mobile Applications

PLANNING AND BUILDING A MOBILE PRESENCE

What is the "right" mobile presence for your firm? The answer depends on identify- ing the business objectives, and from these, deriving the information requirements of your mobile presence. The same kind of systems analysis and design (SAD) reasoning described earlier in the chapter is needed for planning and building a mobile presence, although there are important differences.

TABLE 4.10 SYSTEMS ANALYSIS FOR BUILDING A MOBILE PRESENCE

BUSINESS OBJECTIVE	SYSTEM FUNCTIONALITY	INFORMATION REQUIREMENTS
Driving sales	Digital catalog; product database	Product descriptions, photos, SKUs, inventory
Branding	Showing how customers use your products	Videos and rich media; product and customer demonstrations
Building customer community	Interactive experiences; games with multiple players	Games, contests, forums, social sign-up
Advertising and promotion	Coupons and flash sales for slow-selling items	Product descriptions, coupon management, and inventory management
Gathering customer feedback	Ability to retrieve and store user inputs including text, photos, and video	Customer sign-in and identification; customer database

Developing a Mobile Website and Building Mobile Applications

MOBILE PRESENCE: DESIGN CONSIDERATIONS

Designing a mobile presence is somewhat different from traditional desktop website design because of different hardware, software, and consumer expectations.

TABLE 4.11 UNIQUE FEATURES THAT MUST BE TAKEN INTO ACCOUNT IN DESIGNING A MOBILE PRESENCE

FEATURE	IMPLICATIONS FOR MOBILE PLATFORM
Hardware	Mobile hardware is smaller, and there are more resource constraints in data storage and processing power.
Connectivity	The mobile platform is constrained by slower connection speeds than desktop websites have.
Displays	Mobile displays are much smaller and require simplification. Some screens are not good in sunlight.
Interface	Touch-screen technology introduces new interaction routines that are different from the traditional mouse and keyboard. The mobile platform is not a good data entry tool but can be a good navigational tool.

Developing a Mobile Website and Building Mobile Applications

CROSS-PLATFORM MOBILE APP DEVELOPMENT TOOLS

- There are hundreds of low-cost or open source app development toolkits that make creating cross-platform mobile apps relatively easy and inexpensive without having to use a device-specific programming language.
- For instance, **Flutter** is an open-source toolkit developed by Google that can be used to create native apps for Android and iOS devices (as well as applications for Windows, Mac, and the Web). **React Native** is another open-source tool that uses React and JavaScript to build native iOS, Android, Windows, and Mac apps. **Appery.io** is a cloud-based platform that enables you to use a drag-and-drop visual builder tool to create HTML5 apps using jQuery Mobile. Appery.io supports Android and iOS applications.
- On the more technical side, **PhoneGap** is a mobile development framework that uses software called **Apache Cordova** to enable building hybrid mobile applications using HTML, CSS, and JavaScript.

Developing a Mobile Website and Building Mobile Applications

MOBILE PRESENCE: PERFORMANCE AND COST CONSIDERATIONS

- If you already have a website that you don't want to totally redevelop, the least expensive path is to resize it to create a smartphone-friendly **mobile site**.
- Building a **mobile web app** that uses the mobile device's browser requires more effort and cost than developing a mobile website and suffers from the same limitations as any browser-based application.
- The most expensive path to a mobile presence is to build a **native app**. Native apps can require more programming expertise, although there are many new development packages that can build mobile native apps with minimal programming expertise.

Case Study : Skyscanner - The One-Stop Travel Platform

เรียนรู้ E-Commence Presence ของ Skyscanner และการใช้ประโยชน์ของ E-Commerce ในการแข่งขันกับคู่แข่งธุรกิจ

Do you have any questions?

มีคำถามหรือข้อสงสัยเกี่ยวกับเนื้อหาที่เรียนในวันนี้

Next Session Topic

หัวข้อสำหรับสัปดาห์หน้า

- E-Commerce Security and Payment Systems